



Kampus
Merdeka
INDONESIA JAYA

INTRODUCTION TO MODELING AND SIMULATION

Elly Warni, S. T., M. T.





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Agenda



Lecture contract
(Schedule, RPS, etc.)



Overview of the course



Basic concepts of
Modeling and Simulation



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Lecture Contract

- Quizz (11%)
- Reports (17%)
- Project (40%)
- Participatory (10%)
- Assignment (22%)

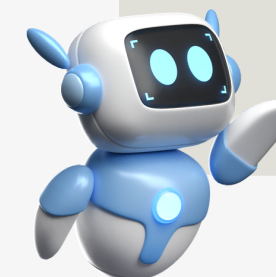


Definition Modeling and Simulation



From the example, Modelling and simulation (M&S) can be used to create computer models of the system, allowing hospital managers to simulate various scenarios to find the causes of long waiting times and test effective reduction strategies.

“Simulation is an attempt to mimic the factual processes of a real environment or a system over a period of time.”





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Model

In this simulation activity, either manually or through a computer process will create/produce a picture of changes or changes in the system against time variables.

So that in these observations will be able to conclude changes in characteristics that can occur in a real system



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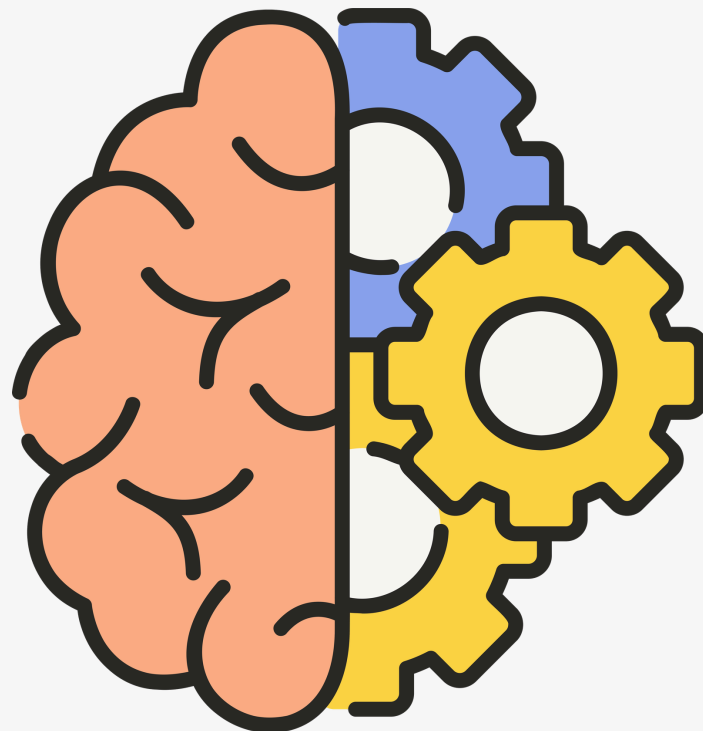
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Logic

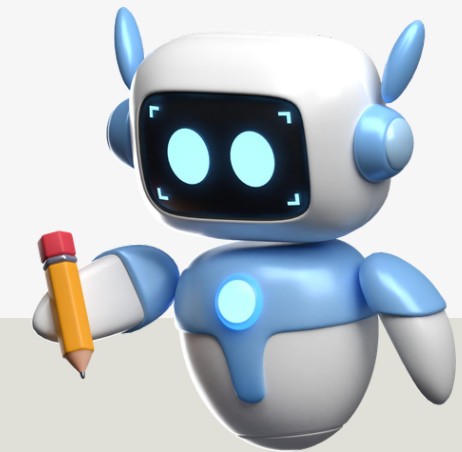
This conceptual form will give birth to a simulation model concept.

Simulation model is usually formed using a symbol or logic statement which will involve assumptions that consider the nature/characteristics/relationships between entities between objects involved which will later describe how the imitated system works.

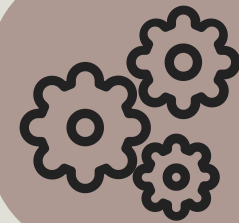
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Definition Modeling and Simulation

By studying the interrelationships and changes of the variables that can affect the system, the right parameters can be determined before the actual system is implemented.



Model



Conceptual form of a system

Simulation



Computational activity on the model to be built



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Goals of Modeling and Simulation



Validation is key to ensuring the accuracy of the simulation model.



Validated model can be used to explore 'what if' scenarios with confidence



Simulation allows us to predict and manage the impact of changes in real systems



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**When the
simulation is the
appropriate tool?**





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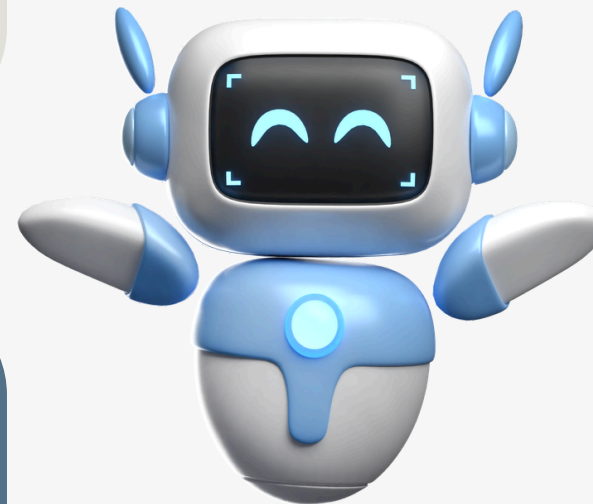
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Simulation enable the study of internal interaction of a subsystem with complex system

2

Informational, organizational and environmental changes can be simulated and fine their effect



3

A simulation help us to gain knowledge about improvement of system

4

Finding important input parameters with changing parameter input



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Simulation can be used with new design and policies before implementation

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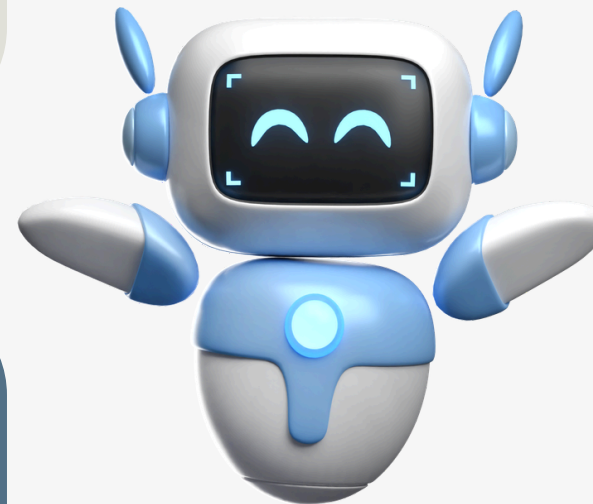
Simulation models design for training make learning possible without cost disruption

6

Simulating different capabilities for a machine can help determine the requirement

8

The modern system (factory, wafer fabrication plant, service organization) is too complex that its internal interaction can be treated only by simulation





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1777- Buffon's Neddle

The origins of M&S can be traced back to Buffon's needle experiments in 1777, which utilised the Monte Carlo method.





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1812- Buffon-Laplace Neddle

Pierre-Simon Laplace later improved on this experiment in 1812, known as the Buffon-Laplace needle problem





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During World War II

During World War II, there were great advances in simulation and computing technology.

The development of computer simulations was used for strategic planning and analysis of military operations.





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In the 1950's

M&S techniques began to be widely applied in industrial systems analysis.





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1970-1980 Evolution of M&S Engineering and Technology

The development of specialised simulation software for various industries began to emerge





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Phase in a Simulation Study

A series of phases in the lifecycle that assist in the modelling and analysis of complex systems.

The cycle is iterative, allowing for continuous revision and development.



Phase in a Simulation Study

Problem Formulation

- Identification of the Problem
- Understanding the System
- Alternative Solutions
- Risk Analysis

Conceptual Model

- Building the Conceptual Model
- State Variables and Relationships
- Critical Features
- Documentations

Input/Output Data

- Data Collection
- Data Classification
- Dealing with Nonexistent Data



Phase in a Simulation Study

Detailed Modeling

- Model Development
- Model Specification
- Assumptions and Simplifications

Simulation Implementation

- Model Implementation
- Simulation Model Construction
- Execution Framework (EF)

Verification and Validation

- Verification
- Validation
- Refinement



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Phase in a Simulation Study

Experimentation

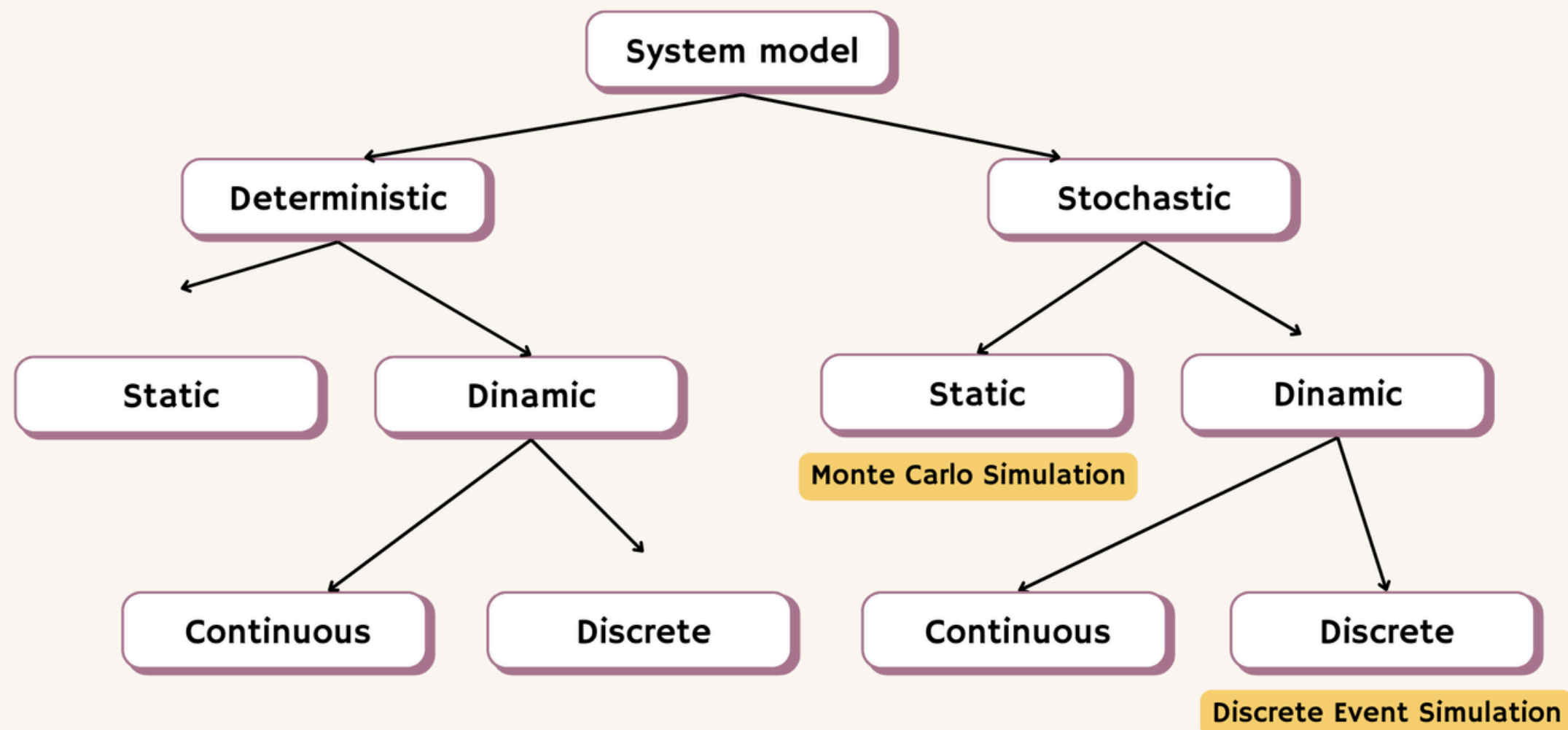
- Designing Experiments
- Execution and Analysis

Output Analysis

- Visualization Tools
- System Behavior Insights

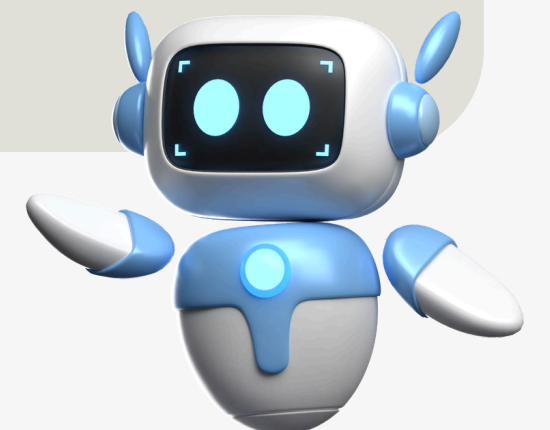


System Model



System

A system is a collection of several objects that are interrelated with one another and will interact with each other towards a certain infrastructure or goal





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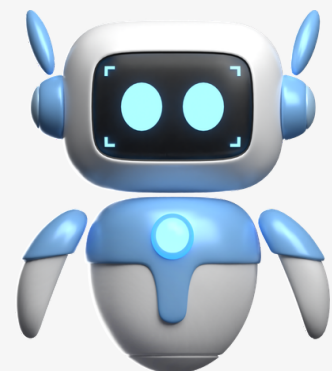
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Deterministic

Models that do not have random variable inputs. Used in situations where all variables can be controlled and predicted

~~random
variable~~





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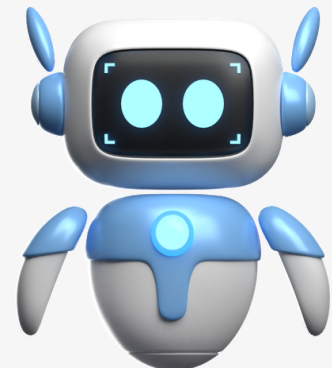
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Stochastic

Stochastic models include an element of chance or uncertainty. The outcome is uncertain, but statistically predictable.

Used in situations where there is uncertainty or variability, such as in queuing theory or finance.





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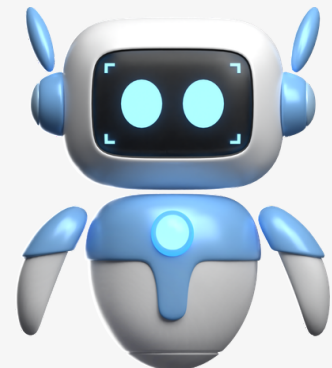
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Static

Static models describe the system at a single point in time, without considering changes over time.

Useful in situations where system conditions can be considered constant over a period of time.





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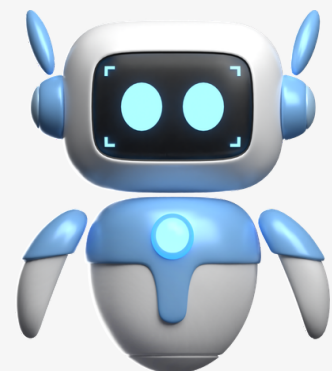
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Dynamic

The dynamic model considers the changes in the system over time.





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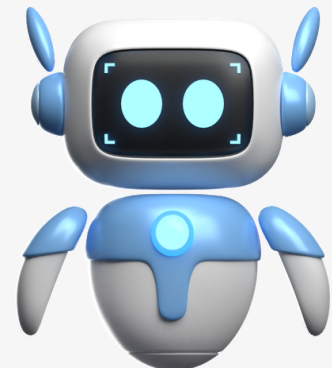
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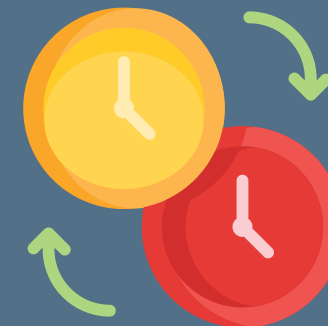
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Continuous

A system in which the state variables change continuously over time. The main reference is time.





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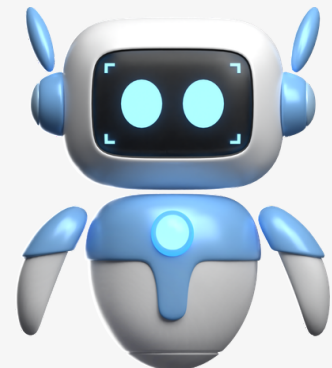
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Discrete

Discrete models describe system changes that occur at specific and discrete points in time.

Based on an event, where event will change the system.





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Types of Simulations





Monte Carlo Simulation



A simulation technique that uses random numbers and mathematical techniques to predict possible outcomes of an uncertain event.

This technique is widely used in various fields to perform estimations and statistical hypotheses test.



Discrete Event Simulation



Discrete Event Simulation (DES) is a modeling approach that focuses on representing the operations of a system as a sequence of discrete events.

Each event occurs at a specific point in time and affects the state of the system



Agent Based Modeling



A simulation technique where individual "agents" act and interact based on predefined rules.

Agents can represent people, groups, or entities within a system, and their interactions result in the dynamics of the overall system.

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Environmental Modeling



The process of building models that represent processes and interactions in the natural environment.

These models are used to understand the impact of environmental changes, forecast outcomes from interventions, and aid in making sustainable policy decisions.



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Examples of problems and their modelling and simulation



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Risk Investment



Monte Carlo Simulation to assess the risk of an investment portfolio. We can simulate the future value of an investment based on historical returns.

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Hospital Queues



Hospitals use DES to simulate patient flow through emergency departments, outpatient clinics, and surgical units. Events include patient arrivals, service completions, and resource availability. The simulation aids in optimizing staffing, reducing wait times, and enhancing patient care

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Market Dynamics



In economic models, agents can represent consumers, firms, or traders, each with different strategies and preferences. ABS helps study market dynamics, competition, and the emergence of economic patterns.



THANK YOU!

See you in The Next Topic!

References

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